An Adaptive Safe Reinforcement Learning Strategy for Multi-Agent Switched Systems

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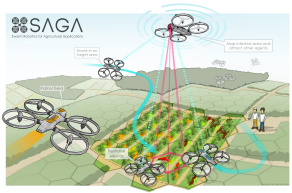
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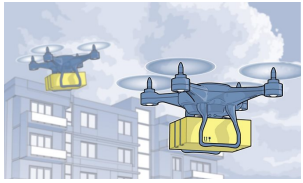
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Motivations and problem to solve

- Safety remains a major challenge in domains where **violations of operational constraints** may lead to **instability** or **irreversible damage**.
- **Safe RL** approaches focus either on centralised low-level control or single-agent frameworks.
- Decentralised **MPC and RL schemes** aim to maintain **interaction safety** while keeping **per-agent optimisation** tractable.
- In particular, our proposal presents a **hierarchical** framework integrating **SMPC** and **dec-MARL**.



Agricultural swarm applications.



Delivery UAVs.

Outline

1. Preliminaries and problem statement
2. Proposed strategy
3. Case study
4. Conclusions

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Preliminaries

Let us introduce the switching multi-agent constrained MDP (SMA-CMDP) framework with associated safe set $\mathcal{C}_i$, i.e.,

$$\mathcal{M} = (\{\mathcal{S}_i\}_{i=1}^M, \{A_i\}_{i=1}^M, \{f_{\mathrm{d}\sigma_i}\}_{i=1}^M, \{r_{h,i}\}_{i=1}^M, \mu, \gamma, \{g_i\}_{i=1}^M), \quad \mathcal{C}_i = \{s_{t,i} \in \mathcal{S}_i : g_i(s_{t,i}) \geq 0\}$$

By assuming that agents learn independently and share a common action space, the SMA-CMDP can be recast as a group of parallel MDPs.

$$\mathcal{M} = \{(\mathcal{S}_i, A_i, f_{\sigma_i}, \{r_{h,i}\}_{h=1}^H, \mu, \gamma, g_i)\}_{i=1}^M, \quad \mathcal{C} = \bigtimes_i \mathcal{C}_i = \{(s_{t,1}, ..., s_{t,M}) \in \mathcal{S} : g_i(s_{t,i}) \geq 0, \forall i \in [M]\}.$$

Adaptive single-agent CBF:

For any given agent $i \in [M]$ with its safe set $\mathcal{C}_i$, the function $g_i(\cdot)$ is a *discrete-time adaptive control barrier function* if $\forall s_{t,i} \in \mathcal{C}_i$, there exists $\alpha(s_{t,i}) \in [0,1]$ such that

$$\sup_{a_{t,i} \in A_i} [g_i(f_{\mathrm{d}\sigma_i}(s_{t,i}, a_{t,i})) + (\alpha(s_{t,i}) - 1) g_i(s_{t,i})] \geq 0.$$

Preliminaries: Quadrotor UAV switched model

Quadrotor UAV kinematics:

$$\mathbf{v} = R_{\mathrm{v}}\mathbf{v_B},\ \ \mathbf{w} = R_{\mathrm{w}}\mathbf{w_B}.$$

N_{r} quadrotors, indexed by $\mathcal{N} \coloneqq \{i \mid i = 1, \ldots, N_{\mathrm{r}}\}$, operating under $\boldsymbol{\sigma}^{[i]} \in \mathcal{M} \coloneqq \{1, \ldots, m\}$ (**switching signal**), which defines the **control input**:

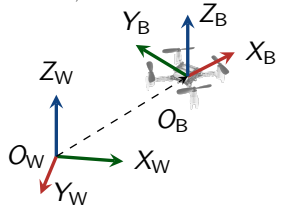
$$\boldsymbol{a_i(\sigma_i)} = \begin{cases} 0_{6\times 1}, & \boldsymbol{\sigma}^{[i]} = 1 \\ \bar{v}e_{\sigma^{[i]}-1}, & \boldsymbol{\sigma}^{[i]} = 2,3,4 \\ \bar{w}e_{\sigma^{[i]}-1}, & \boldsymbol{\sigma}^{[i]} = 5,6,7 \\ -\bar{v}e_{\sigma^{[i]}-7}, & \boldsymbol{\sigma}^{[i]} = 8,9,10 \\ -\bar{w}e_{\sigma^{[i]}-7}, & \boldsymbol{\sigma}^{[i]} = 11,12,13 \end{cases}.$$

Continuous-time **switched model** for UAV i:

$$\dot{x}_i = f_{\sigma_i}(x_i, a_i(\sigma_i)).$$

Euler discretization method with sampling time $T > 0$ and sampling instant k:

$$s_{k+1,i} = s_{k,i} + Tf_{\sigma_{k,i}}(s_{k,i}, a_{k,i}(\sigma_{k,i})),$$
$$s_{k+1,i} = f_{\mathrm{d}\sigma_{k,i}}(s_{k,i}, a_{k,i}(\sigma_{k,i})).$$



Problem statement

Consider the **aggregation problem** for a multi-agent system of M **quadrotor UAVs**, each governed by **switching dynamics**, such that each agent is capable of synthesising its own optimal control action under a **decentralised network structure**.

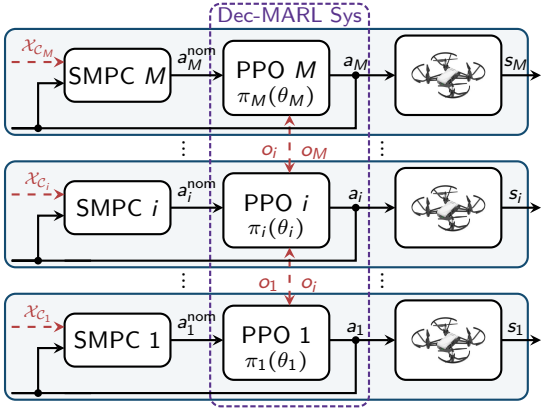
Moreover, the agent trajectories are required to be **collision free** while keeping the **state in a safe manoeuvrability set**. Finally, the final aggregated configuration must occur in the vicinity of a predefined reference position.

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The proposed control scheme

- In multi-agent settings, computational burden of common centralised configurations increases with the number of agents.
- A hierarchical scheme with a decentralised safety control structure is proposed, combining SMPC and dec-MARL.
- In particular, the optimal solutions found by the SMPCs (the nominal actions $a_{t,i}^{\text{nom}}$) are fed into their corresponding decentralised MARL controller.
- The dec-MARL controller evaluates the nominal action safety based and proposes a safe action in case of constraint violations.



Hierarchical control scheme.

Distributed switched MPC

- Computational load is divided by adopting a **distributed cooperative scheme**.
- Each UAV solves an SMPC using a **cluster-based topology**. The cluster membership for UAV i is

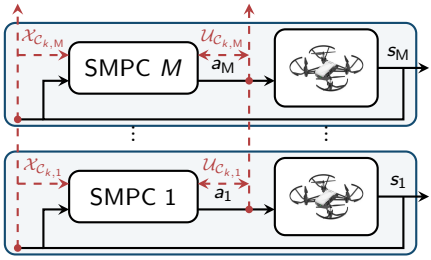
$$\mathcal{C}_{k,i} \coloneqq \{j \in \mathcal{N} \mid \|p_{k,i} - p_{k,j}\|_2 < r_{\text{cl}},\ j \neq i\}.$$

- The shared information is collected in

$$\mathcal{X}_{\mathcal{C}_{k,i}} \coloneqq \{s_{k,j} \in \mathbb{R}^6 \mid j \in \mathcal{C}_{k,i}\},$$
$$\mathcal{U}_{\mathcal{C}_{k,i}} \coloneqq \{\mathbf{u}_{k,j} = [\hat{u}_{k+1,j}, \ldots, \hat{u}_{k+N-1,j}] \mid j \in \mathcal{C}_{k,i}\}.$$

- The FHOCP is

$$\min_{\boldsymbol{\sigma}_{k,i}} \mathcal{V}(s_{k,i}) = \frac{1}{|\mathcal{X}_{\mathcal{C}_i}|} \sum_{j \in \mathcal{X}_{\mathcal{C}_i}} \|s^{\text{lp}}_{k,i} - s^{\text{lp}}_{k,j}\|_2^2 - \|s^{\text{lp}}_{k,i} - p_{\text{ref}}\|_2^2,$$
$$\text{s.t.} \quad s_{p+1,i} = f_{\text{d}_{\sigma_{p,i}}}(s_{p,i}, a(\sigma_{p,i}))$$
$$s_{p,i} \in \mathcal{X}_p, \quad \forall p \in \mathcal{T}_k \cup \{k+N\}.$$



Sequential distributed SMPC scheme.

Safe RL control via PPO-Lagrangian

Nominal policy optimisation problem

$$\max_{\theta_i} \mathcal{J}_{\mathrm{R}}(\theta_i) = \mathbb{E}_{\pi_{\theta_i}}\left[\sum_{t}^{H} \alpha^t r_{t,i}(s_{t,i}, a_{t,i}, s_{t+1,i})\right]$$

$$\text{s.t.} \quad \mathcal{J}_{\mathrm{C}}(\theta_i) = \mathbb{P}_{\pi_{\theta_i}}\begin{bmatrix} \|s^{\mathrm{lp}}_{t,i} - s^{\mathrm{lp}}_{t,j}\|_2^2 > d^2_{\mathrm{safe}} \\ \phi^2_{t,i} > \phi^2_{\mathrm{safe}} \\ \nu^2_{t,i} > \nu^2_{\mathrm{safe}} \end{bmatrix} \leq l_{\mathrm{tol}},$$

Lagrangian relaxation

Safety requirements are formulated into the learning process by introducing a dual variable $\lambda_i \geq 0, \ \forall i \in [M]$, such that the nominal problem turns unconstrained, i.e.,

$$\min_{\lambda_i \geq 0} \max_{\theta_i \in \Theta} \ \mathcal{J}(\theta_i, \lambda_i) = \min_{\lambda_i \geq 0} \max_{\theta_i \in \Theta} \ \mathcal{J}_{\mathrm{R}}(\theta_i) - \lambda_i \mathcal{J}_{\mathrm{C}}(\theta_i),$$

Proximal policy optimisation

- In dec-MARL, per-agent policy optimisation in the actor network is performed via gradient ascent over the advantage function $A^{\pi_i(\theta_i)}(s_i, a_i)$.
- In the actor network, the gradient is approximated using a smooth generalised advantage estimation (GAE) and the importance sampling (IS) ratio, i.e.,

$$A_{h,i}^{\pi_i(\theta_i)}(s_{h,i}, a_{h,i}) \approx \sum_{l=0}^{\infty} (\gamma\beta)^l \delta_{h+l,i}, \quad r_{t,i}(\theta_i) = \frac{\pi_i(a_{t,i}|s_{t,i}, \theta_i)}{\pi_i^{\text{old}}(a_{t,i}|s_{t,i}, \theta_i)}$$

- A separate critic network V_{ϕ_i} is built for each agent to learn its value function, enhancing learning convergence.
- The Lagrangian term is embedded into the PPO objective to ensure update stability over the episodes.

$$\mathcal{L}_i(\theta_i, \lambda_i) = \mathbb{E}_h \left[\min \left(r_{t,i}(\theta_i) A_{h,i}^{\text{R}}, \text{clip}(r_{h,i}(\theta_i), 1-\epsilon, 1+\epsilon) A_{h,i}^{\text{R}} \right) \right] - \lambda_i A_{h,i}^{\text{C}}.$$

Environment model

- The environment models the cooperative coordination of $M = 4$ quadcopters in a bounded cubic workspace with a side length of $4\,\mathrm{m}$ with sampling time $T = 0.5\,\mathrm{s}$.
- Each agent is modelled as a sphere of radius $r_b = 0.05\,\mathrm{m}$ and evolves according to switched nonlinear kinematics.
- The reward expression discourages the dispersion of the agents, while pulling the collective centre of mass towards the target.

$$r_{h,i}(s_{h,i}) = -\frac{\mathrm{w}_{r,\mathrm{j}}}{|\mathcal{X}_{\mathcal{C}_i}|} \sum_{s^{\mathrm{lp}}_{h,j} \in \mathcal{X}_{\mathcal{C}}^{[i]}} \|s^{\mathrm{lp}}_{h,i} - s^{\mathrm{lp}}_{h,j}\|_2^2 - \mathrm{w}_{r,\mathrm{o}} \|s^{\mathrm{lp}}_{h,i} - p_{\mathrm{ref}}\|_2^2,$$

- Safety is defined through the sum of two contributions associated with the distance between agents and the roll and pitch angle relative errors.

$$g(s_i) = \overbrace{\mathrm{w}_{\mathrm{CBF}} \sum_{j \in \mathcal{X}_{\mathcal{C}}^{[i]}} \max(0,\, d_{\mathrm{safe}} - \|s^{\mathrm{lp}}_i - s^{\mathrm{lp}}_j\|_2^2)} + \underbrace{\mathrm{w}_{\phi,\nu} \left(\max(0, \phi_i^2 - \phi_{\mathrm{safe}}^2) + \max(0, \nu_i^2 - \nu_{\mathrm{safe}}^2)\right)}.$$

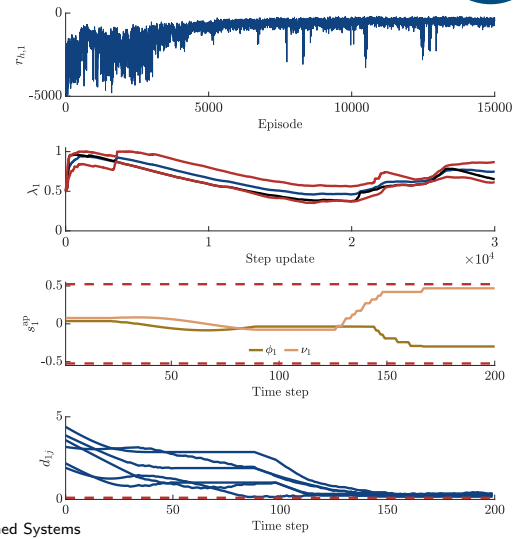
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Policy training

- Per-agent policy optimisation along 15000 episodes.
- Evaluation of the adaptive response of the Lagrange parameters across each episode.
- The resultant policies were evaluated starting from a random initial configuration for $H = 200$ time steps.

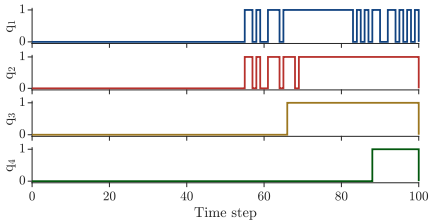
γ	α_{θ_i}	ϵ	β	$\lambda_{0,i}$	α_{λ_i}	$\mathbf{w}_{r_{i,j}}$	$\mathbf{w}_{r_{i,o}}$	$\mathbf{w}_{\mathrm{CBF}}$	$\mathbf{w}_{\phi,\nu}$
0.98	10^{-4}	0.25	0.97	0.5	5.10^{-4}	1	2	10	10

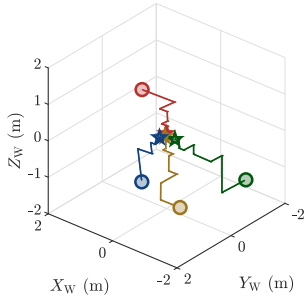


Simulation outcome

- The case study simulation uses a prediction horizon $N = 5$ for $H = 100$ time steps, starting from a **random initial configuration**.
- The **binary indicator** q_i returns 1 if the nominal action is overridden with a **safe action** produced by the MARL policy, otherwise 0.
- A key comparison between our proposal and the baseline strategy showed a **reduction of one order of magnitude** and a **ready-to-implement computational time**.

Agents successfully aggregate and avoid collisions among them!





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Conclusions and future work

- This work introduces a decentralised SMA-CMDP framework for the safe aggregation problem in a multi-UAV system.
- The proposed approach relaxes the baseline strategy, decoupling safe constraint validation from the SMPC controllers and driving it with decentralised MARL PPO-based policies in a hierarchical fashion.
- The training phase outcome exhibits a clear convergence behaviour of the cumulative rewards, anticipating the correct action management obtained when implemented in the overall control scheme.
- Future work will study the behaviour of the framework in more challenging applications, e.g., large-scale formations with dynamic obstacles, Crazyflie UAVs implementation, and the impact of the hyperparameters.

Thank you for the attention!

